Lab-7: JavaScript:2 (Events and HTML Document Object Model)

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>JavaScript Exercises 1-8</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 20px; }

    .tabs { display: flex; gap: 5px; flex-wrap: wrap; margin-bottom: 20px; }
    .tabs button { padding: 8px 14px; cursor: pointer; background: #ddd; border: 1px solid #aaa; }
    .tabs button.active { background: #333; color: white; }

    .page { display: none; }
    .page.active { display: block; }

    h2 { color: #333; border-bottom: 2px solid #333; padding-bottom: 5px; }

    input, select, button {
      padding: 8px;
      margin: 5px 0;
```

```
    font-size: 14px;
  }

  input, select { border: 1px solid #aaa; width: 250px; }
  button.btn { background: #2c3e50; color: white; border: none; cursor: pointer; width: 260px; }
  button.btn:hover { background: #444; }

  .error { color: red; font-size: 12px; display: none; }
  .error.show { display: block; }

  table { border-collapse: collapse; width: 100%; max-width: 600px; }
  table td, table th { border: 1px solid #ccc; padding: 8px 12px; text-align: left; font-size: 13px; }
  table th { background: #eee; }

  .row { padding: 6px 0; border-bottom: 1px solid #eee; }
  .row label { display: inline-block; width: 200px; font-weight: bold; color: #555; }

  #mouseArea {
    width: 100%;
    max-width: 500px;
    height: 150px;
    background: #f0f0f0;
    border: 1px solid #aaa;
    display: flex;
```

```
        align-items: center;
        justify-content: center;
        cursor: crosshair;
        margin: 10px 0;
      }
  </style>
</head>
<body>

<h1>JavaScript Exercises 1-8</h1>

<!-- TAB BUTTONS -->
<div class="tabs">
  <button class="active" onclick="show('ex1', this)">Ex 1: Button Window</button>
  <button onclick="show('ex2', this)">Ex 2: Location</button>
  <button onclick="show('ex3', this)">Ex 3: Navigator</button>
  <button onclick="show('ex4', this)">Ex 4: Browser Detect</button>
  <button onclick="show('ex5', this)">Ex 5: History</button>
  <button onclick="show('ex6', this)">Ex 6: Change Event</button>
  <button onclick="show('ex7', this)">Ex 7: Mouse & Key</button>
  <button onclick="show('ex8', this)">Ex 8: Form Validate</button>
</div>

<div id="ex1" class="page active">
  <h2>Exercise 1 - Open New Window on Button Click</h2>
  <p>Click the button below to open a new window.</p>
```

```
  <button class="btn" onclick="window.open('https://www.google.com', '_blank',
'width=800,height=600')">
    Click Me - Open New Window
  </button>
</div>

<div id="ex2" class="page">
  <h2>Exercise 2 - Location Object Details</h2>
  <table id="locTable">
    <tr><th>Property</th><th>Value</th></tr>
  </table>
  <script>
    var locProps =
['href','protocol','host','hostname','port','pathname','search','hash','origin'];
    var lt = document.getElementById('locTable');
    locProps.forEach(function(p) {
      var row = lt.insertRow();
      row.insertCell().textContent = p;
      row.insertCell().textContent = window.location[p] || '(empty)';
    });
  </script>
</div>

<div id="ex3" class="page">
  <h2>Exercise 3 - Navigator Object Details</h2>
  <table id="navTable">
```

```
      <tr><th>Property</th><th>Value</th></tr>
    </table>
    <script>
      var navProps = {
        appName: navigator.appName,
        appVersion: navigator.appVersion,
        userAgent: navigator.userAgent,
        platform: navigator.platform,
        language: navigator.language,
        onLine: navigator.onLine,
        cookieEnabled: navigator.cookieEnabled,
        hardwareConcurrency: navigator.hardwareConcurrency,
        maxTouchPoints: navigator.maxTouchPoints
      };
      var nt = document.getElementById('navTable');
      for (var key in navProps) {
        var row = nt.insertRow();
        row.insertCell().textContent = key;
        row.insertCell().textContent = navProps[key];
      }
    </script>
  </div>

  <div id="ex4" class="page">
    <h2>Exercise 4 - Browser Name Detection and Redirect</h2>
    <p id="browserResult"></p>
```

```
<p><b>User Agent:</b> <span id="uaText" style="font-size:12px;"></span></p>
<button class="btn" onclick="redirectBrowser()">Open Browser-Specific Page</button>
<script>
  function detectBrowser() {
    var ua = navigator.userAgent;
    if (ua.indexOf('Edg') > -1)   return 'Edge';
    if (ua.indexOf('OPR') > -1)   return 'Opera';
    if (ua.indexOf('Firefox') > -1) return 'Firefox';
    if (ua.indexOf('Chrome') > -1)  return 'Chrome';
    if (ua.indexOf('Safari') > -1)  return 'Safari';
    return 'Unknown';
  }
  var browser = detectBrowser();
  document.getElementById('browserResult').innerHTML = '<b>Detected Browser:</b> '
+ browser;
  document.getElementById('uaText').textContent = navigator.userAgent;


  function redirectBrowser() {
    var pages = {
      Chrome:  'chrome_version.html',
      Edge:   'edge_version.html',
      Firefox: 'firefox_version.html',
      Safari:  'safari_version.html',
      Opera:   'opera_version.html',
      Unknown: 'default_version.html'
    };
```

```
        alert('Redirecting to: ' + pages[browser]);
        // window.location.href = pages[browser];
      }
  </script>
</div>


<div id="ex5" class="page">
  <h2>Exercise 5 - History Object Details</h2>
  <div class="row"><label>history.length</label> <span id="hlen"></span></div>
  <div class="row"><label>history.scrollRestoration</label> <span
id="hscroll"></span></div>
  <div class="row"><label>history.state</label> <span id="hstate"></span></div>
  <br>
  <button class="btn" onclick="history.back()">Back</button><br>
  <button class="btn" onclick="history.forward()">Forward</button><br>
  <button class="btn" onclick="history.go(-1)">Go(-1)</button><br>
  <button class="btn" onclick="pushNewState()">Push State</button>
  <script>
    function refreshHistory() {
        document.getElementById('hlen').textContent    = history.length;
        document.getElementById('hscroll').textContent = history.scrollRestoration || 'N/A';
        document.getElementById('hstate').textContent  = JSON.stringify(history.state) ||
'null';
    }
    function pushNewState() {
        history.pushState({ page: 'demo' }, 'Demo', '#demo');
```

```
        refreshHistory();
      }
      refreshHistory();
    </script>
</div>

<div id="ex6" class="page">
    <h2>Exercise 6 - Change Event with Selection List</h2>

    <label>Item:</label><br>
    <select id="itemSelect" onchange="updateModels()">
      <option value="">-- Select Item --</option>
      <option value="Monitor">Monitor</option>
      <option value="HDD">HDD</option>
      <option value="Keyboard">Keyboard</option>
    </select>
    <br><br>

    <label>Model:</label><br>
    <select id="modelSelect" onchange="showSelected()">
      <option value="">-- Select Model --</option>
    </select>
    <br><br>

    <div id="selResult"></div>
```

```
<script>
  var modelData = {
    Monitor:  ['Acer 21', 'Dell 22', 'LG 24'],
    HDD:     ['500GB', '1TB', '2TB'],
    Keyboard: ['Logitech K120', 'Dell KB216']
  };

  function updateModels() {
    var item = document.getElementById('itemSelect').value;
    var modelSel = document.getElementById('modelSelect');
    modelSel.innerHTML = '<option value="">-- Select Model --</option>';
    document.getElementById('selResult').innerHTML = '';
    if (item && modelData[item]) {
      for (var i = 0; i < modelData[item].length; i++) {
        modelSel.innerHTML += '<option>' + modelData[item][i] + '</option>';
      }
    }
  }

  function showSelected() {
    var item  = document.getElementById('itemSelect').value;
    var model = document.getElementById('modelSelect').value;
    if (item && model) {
      document.getElementById('selResult').innerHTML =
        '<b>Selected Item:</b> ' + item + '<br><b>Selected Model:</b> ' + model;
    }
```

```
    }
  </script>
</div>

<div id="ex7" class="page">
  <h2>Exercise 7 - Mouse Location and Keyboard Key Code</h2>

  <h4>Mouse Position</h4>
  <div id="mouseArea">Move mouse here</div>
  <p><b>X:</b> <span id="mx">0</span>   <b>Y:</b> <span id="my">0</span></p>

  <h4>Keyboard Input</h4>
  <input type="text" id="keyInput" placeholder="Type a key here..." style="width:250px;">
  <p><b>Key:</b> <span id="keyChar">-</span></p>
  <p><b>Key Code:</b> <span id="keyCode">-</span></p>
  <p><b>Unicode:</b> <span id="keyUnicode">-</span></p>

  <script>
    document.getElementById('mouseArea').addEventListener('mousemove', function(e) {
      var rect = this.getBoundingClientRect();
      document.getElementById('mx').textContent = Math.round(e.clientX - rect.left);
      document.getElementById('my').textContent = Math.round(e.clientY - rect.top);
    });

    document.getElementById('keyInput').addEventListener('keydown', function(e) {
      var code    = e.keyCode || e.which;
```

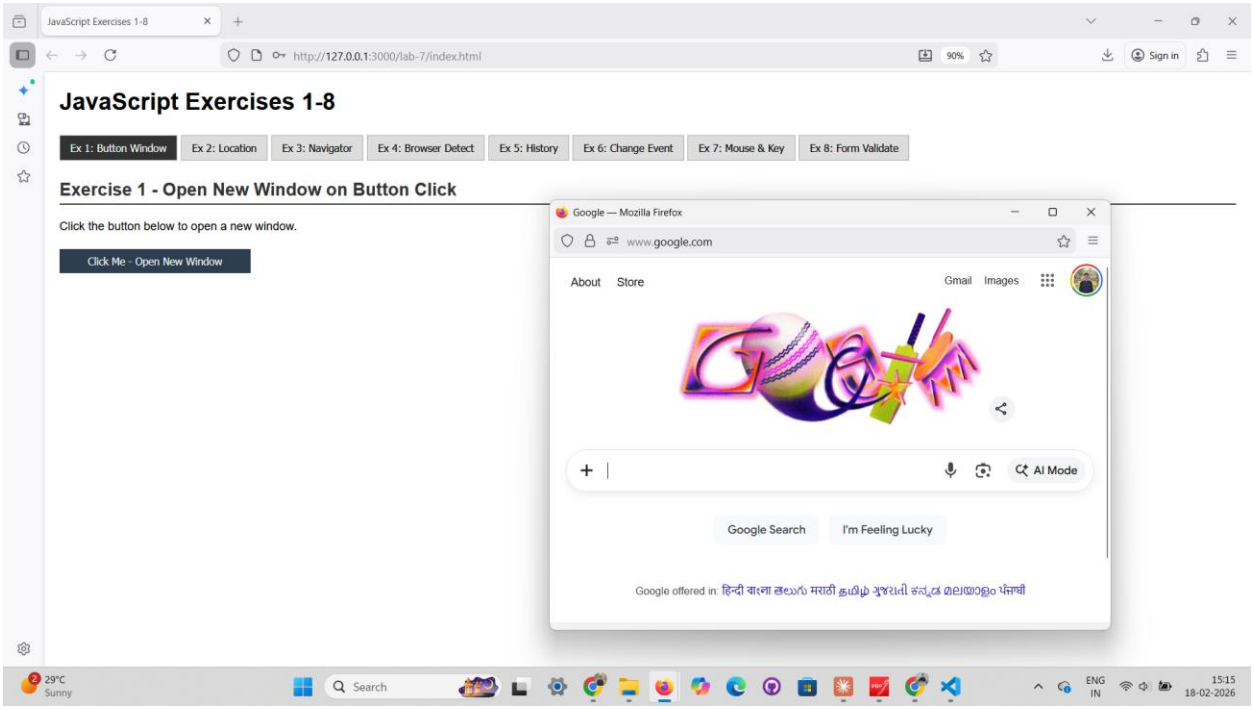
```
        var unicode = e.key.length === 1 ? e.key.charCodeAt(0) : code;
        document.getElementById('keyChar').textContent   = e.key;
        document.getElementById('keyCode').textContent   = code;
        document.getElementById('keyUnicode').textContent = 'U+' +
unicode.toString(16).toUpperCase().padStart(4, '0') + ' (' + unicode + ')';
      });
  </script>
</div>

<div id="ex8" class="page">
  <h2>Exercise 8 - Register Yourself (Form Validation)</h2>

  <label>Full Name *</label><br>
  <input type="text" id="name" placeholder="Enter name"><br>
  <span class="error" id="nameErr">Name is required (letters only, min 3 chars)</span>

  <br>
  <label>Email *</label><br>
  <input type="text" id="email" placeholder="Enter email"><br>
  <span class="error" id="emailErr">Enter a valid email address</span>

  <br>
  <label>Password *</label><br>
  <input type="password" id="password" placeholder="Min 6 characters"><br>
  <span class="error" id="passwordErr">Password must be at least 6 characters</span>
```

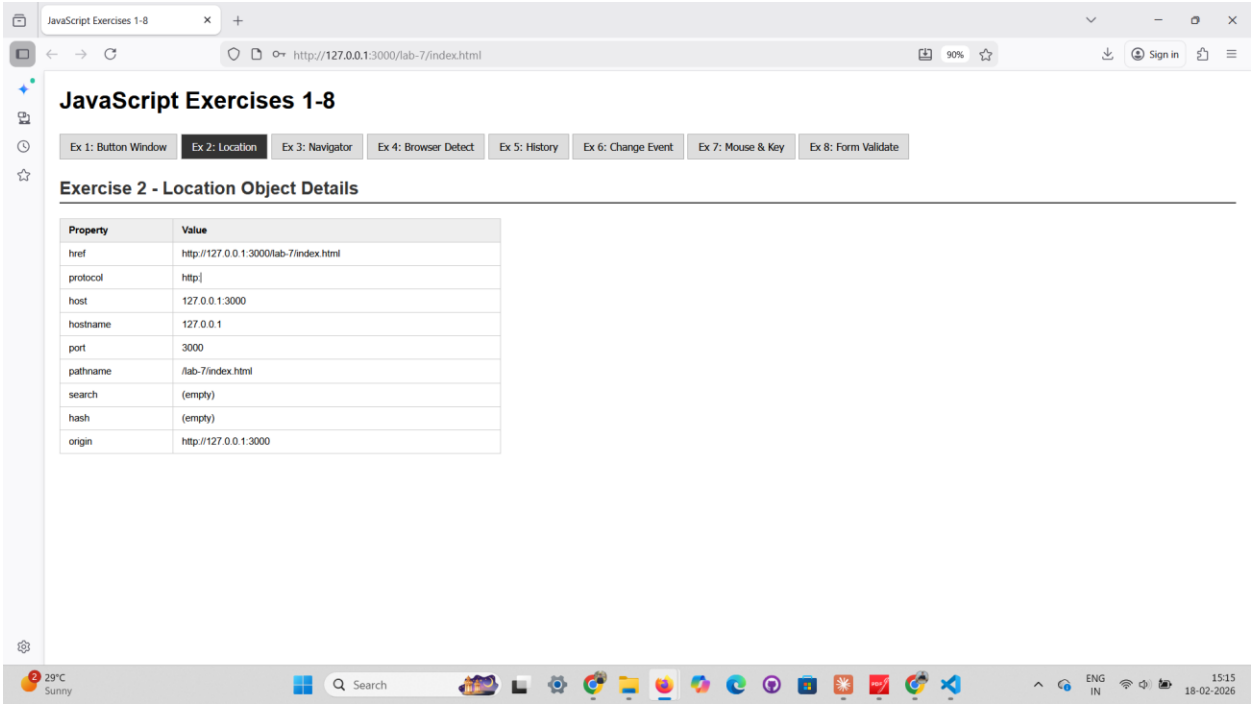
```
<br>
<label>Mobile *</label><br>
<input type="text" id="mobile" placeholder="10-digit number"><br>
<span class="error" id="mobileErr">Enter valid 10-digit mobile number</span>

<br>
<label>Gender *</label><br>
<select id="gender" style="width:268px;">
  <option value="">-- Select Gender --</option>
  <option>Male</option>
  <option>Female</option>
  <option>Other</option>
</select><br>
<span class="error" id="genderErr">Please select a gender</span>

<br><br>
<button class="btn" onclick="validateForm()">Register</button>
<p id="successMsg" style="color:green; display:none;">Registration Successful!</p>

<script>
  function validateForm() {
    var valid = true;

    var name     = document.getElementById('name').value.trim();
    var email    = document.getElementById('email').value.trim();
    var password = document.getElementById('password').value;
```

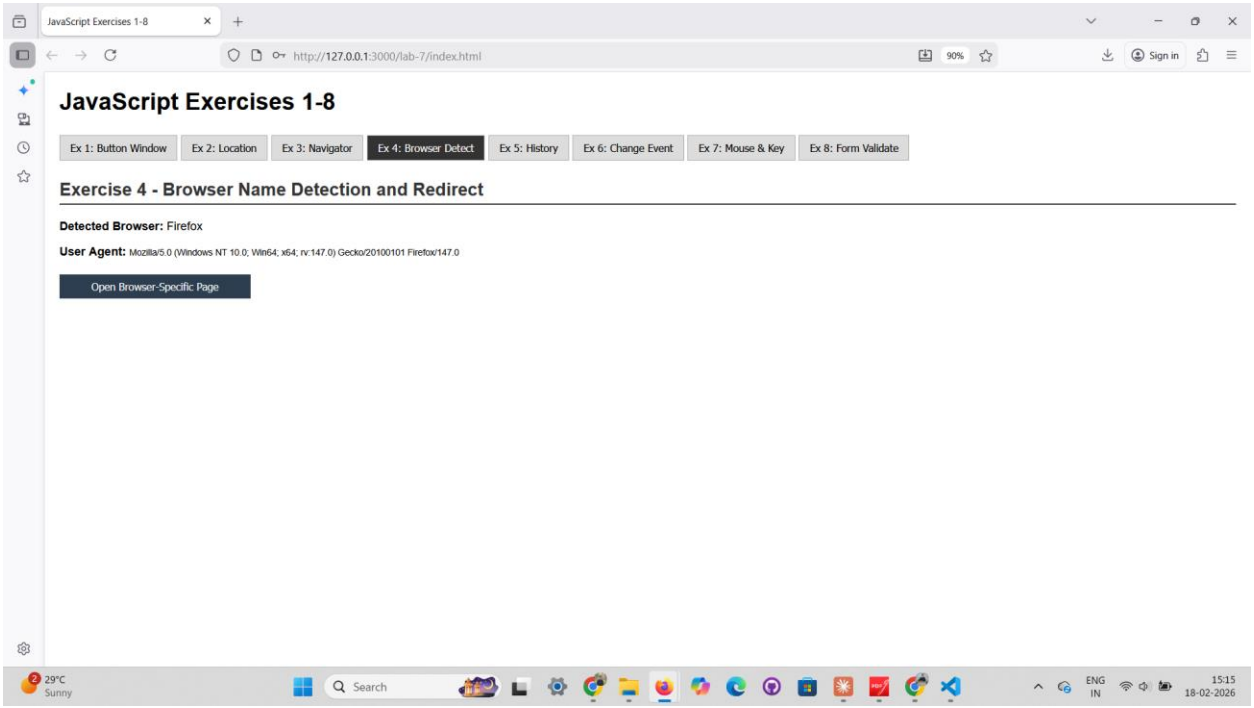
```
var mobile   = document.getElementById('mobile').value.trim();
var gender   = document.getElementById('gender').value;

document.getElementById('successMsg').style.display = 'none';

if (!name) { alert('Name field is blank!'); }
var nameOk = name.length >= 3 && /^[a-zA-Z\s]+$/.test(name);
document.getElementById('nameErr').className    = nameOk ? 'error' : 'error show';

if (!email) { alert('Email field is blank!'); }
var emailOk = /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(email);
document.getElementById('emailErr').className    = emailOk ? 'error' : 'error show';

if (!password) { alert('Password field is blank!'); }
var passOk = password.length >= 6;
document.getElementById('passwordErr').className = passOk ? 'error' : 'error
show';

if (!mobile) { alert('Mobile field is blank!'); }
var mobOk = /^\d{10}$/.test(mobile);
document.getElementById('mobileErr').className   = mobOk ? 'error' : 'error show';

if (!gender) { alert('Please select a gender!'); }
var genOk = gender !== '';
document.getElementById('genderErr').className   = genOk ? 'error' : 'error show';
```

```
        if (nameOk && emailOk && passOk && mobOk && genOk) {
          document.getElementById('successMsg').style.display = 'block';
        }
      }
    </script>
</div>

<script>
    function show(id, btn) {
      var pages = document.querySelectorAll('.page');
      var btns  = document.querySelectorAll('.tabs button');
      for (var i = 0; i < pages.length; i++) pages[i].classList.remove('active');
      for (var i = 0; i < btns.length; i++)  btns[i].classList.remove('active');
      document.getElementById(id).classList.add('active');
      btn.classList.add('active');
    }
</script>

</body>
</html>
```

Output:


JavaScript Exercises 1-8
http://127.0.0.1:3000/lab-7/index.html
JavaScript Exercises 1-8
Ex 1: Button Window
Ex 2: Location
Ex 3: Navigator
Ex 4: Browser Detect
Ex 5: History
Ex 6: Change Event
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 1 - Open New Window on Button Click
Click the button below to open a new window.
Click Me - Open New Window
Google — Mozilla Firefox
www.google.com
About
Store
Gmail
Images
AI Mode
Google Search
I'm Feeling Lucky
Google offered in: हिन्दी বাংলা తెలుగు मराठी தமிழ் ગુજરાતી ಕನ್ನಡ മലയാളം ਪੰਜਾਬੀ


JavaScript Exercises 1-8
http://127.0.0.1:3000/lab-7/index.html
JavaScript Exercises 1-8
Ex 1: Button Window
Ex 2: Location
Ex 3: Navigator
Ex 4: Browser Detect
Ex 5: History
Ex 6: Change Event
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 2 - Location Object Details
Property | Value
href | http://127.0.0.1:3000/lab-7/index.html
protocol | http:
host | 127.0.0.1:3000
hostname | 127.0.0.1
port | 3000
pathname | /lab-7/index.html
search | (empty)
hash | (empty)
origin | http://127.0.0.1:3000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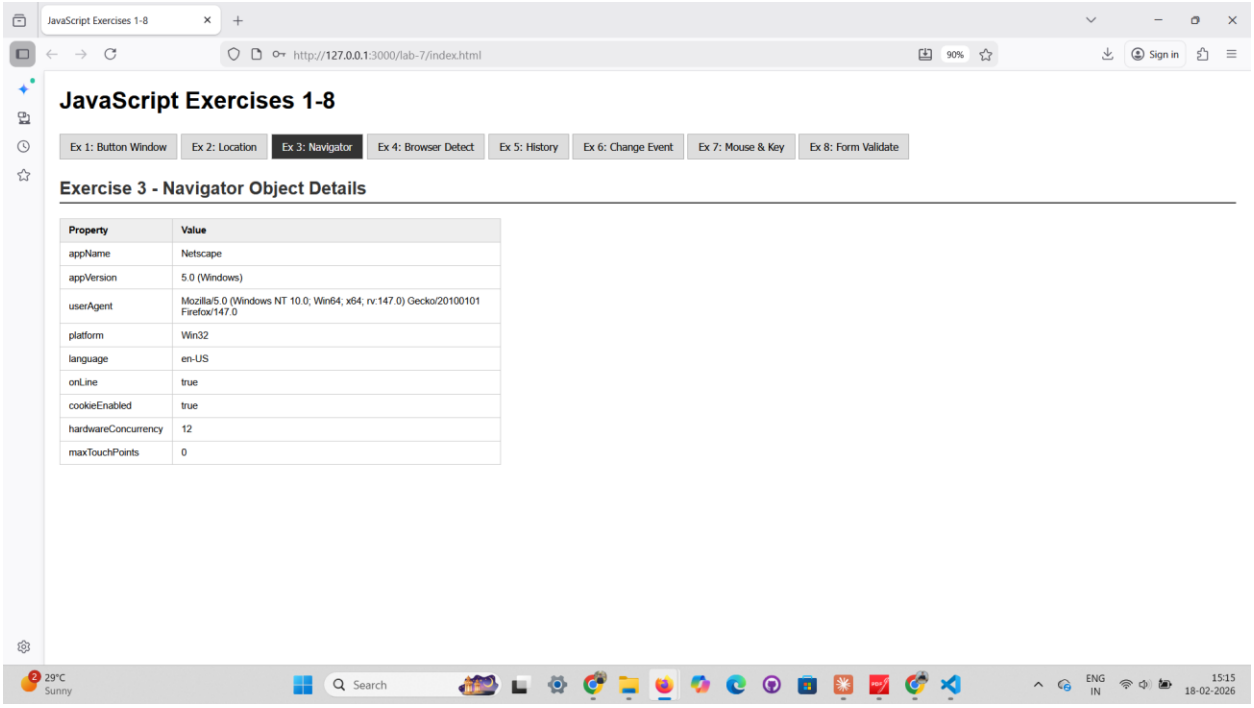
Ex 5: History
Ex 6: Change Event
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 4 - Browser Name Detection and Redirect
Detected Browser: Firefox
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:147.0) Gecko/20100101 Firefox/147.0
Open Browser-Specific Page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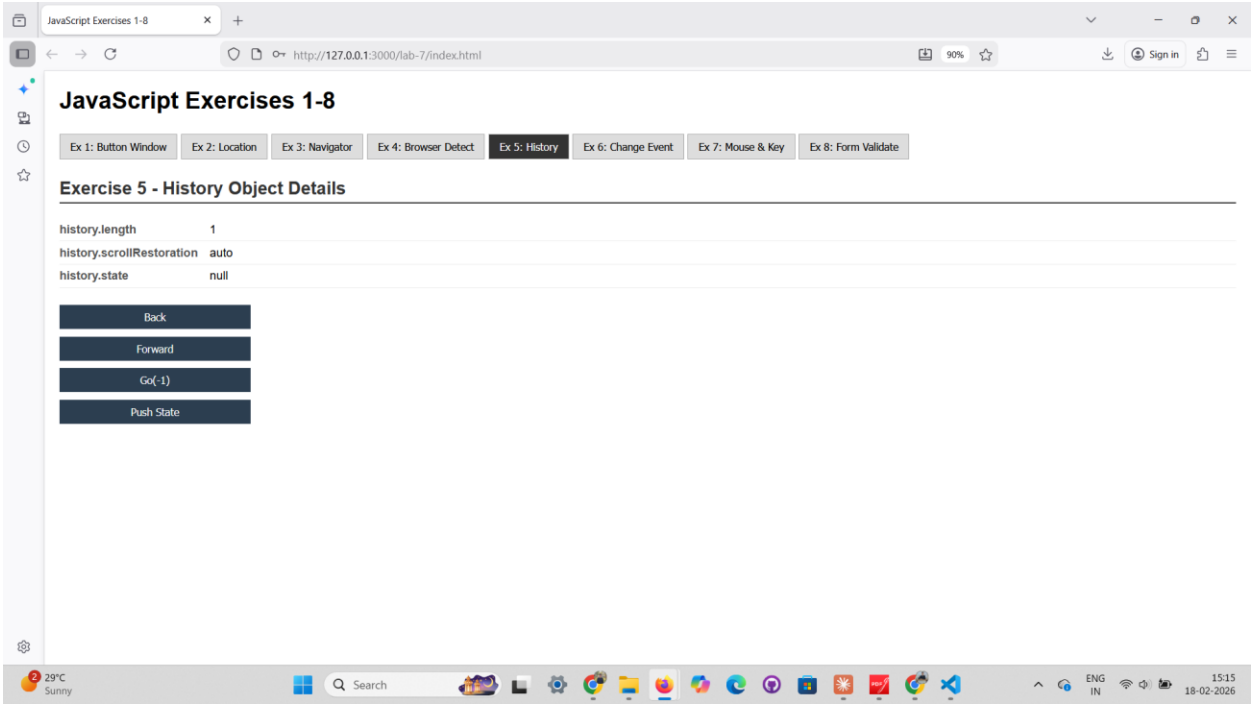
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 3 - Navigator Object Details
Property | Value
appName | Netscape
appVersion | 5.0 (Windows)
userAgent | Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:147.0) Gecko/20100101 Firefox/147.0
platform | Win32
language | en-US
onLine | true
cookieEnabled | true
hardwareConcurrency | 12
maxTouchPoints | 0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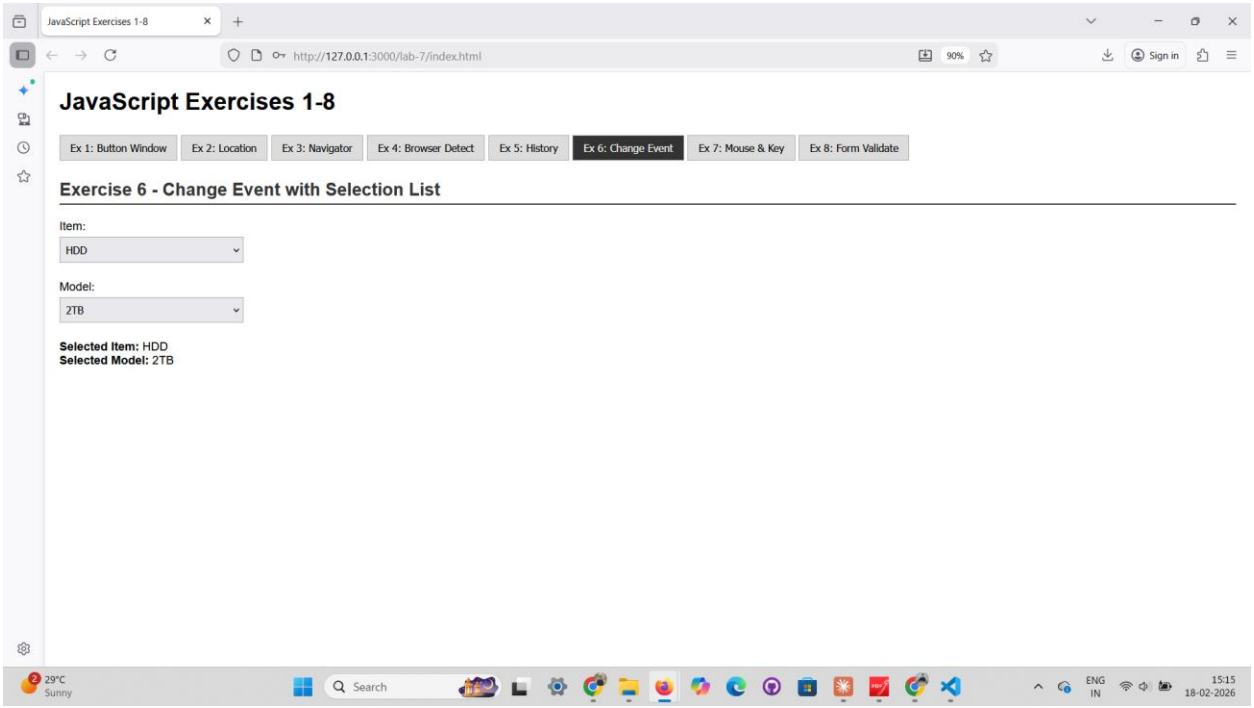
Ex 5: History
Ex 6: Change Event
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 5 - History Object Details
history.length 1
history.scrollRestoration auto
history.state null
Back
Forward
Go(-1)
Push State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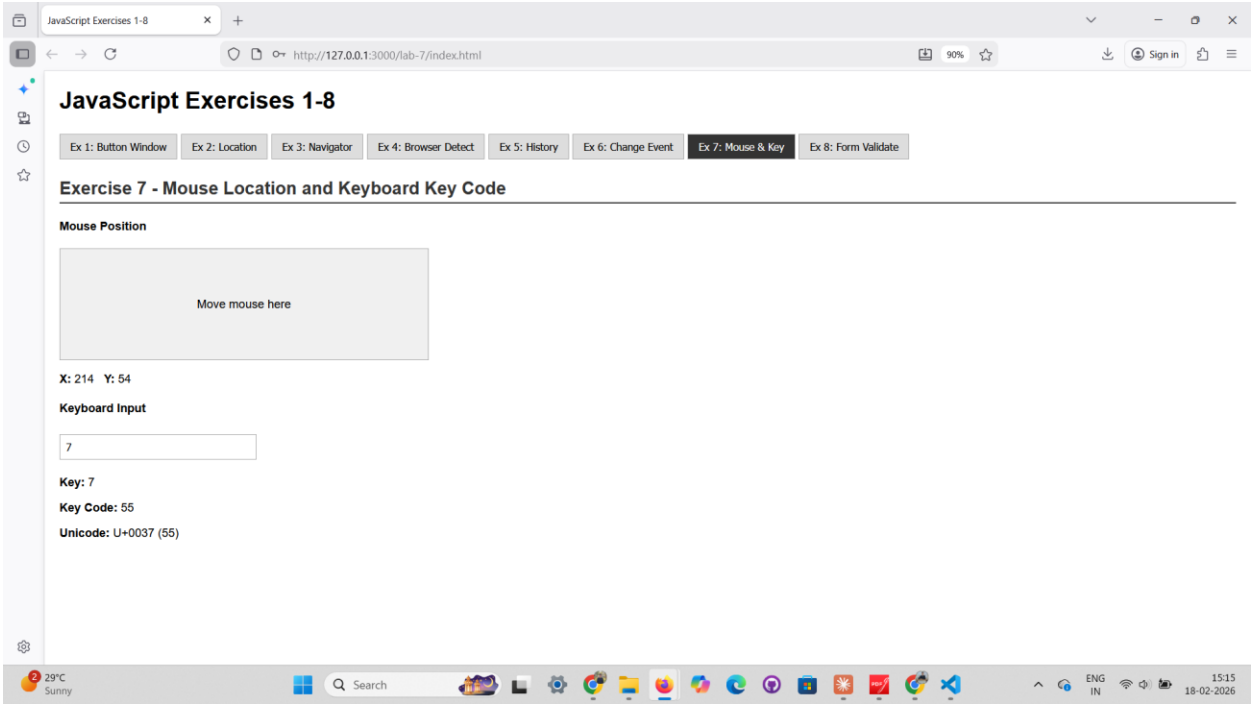
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 6 - Change Event with Selection List
Item:
HDD
Model:
2TB
Selected Item: HDD
Selected Model: 2TB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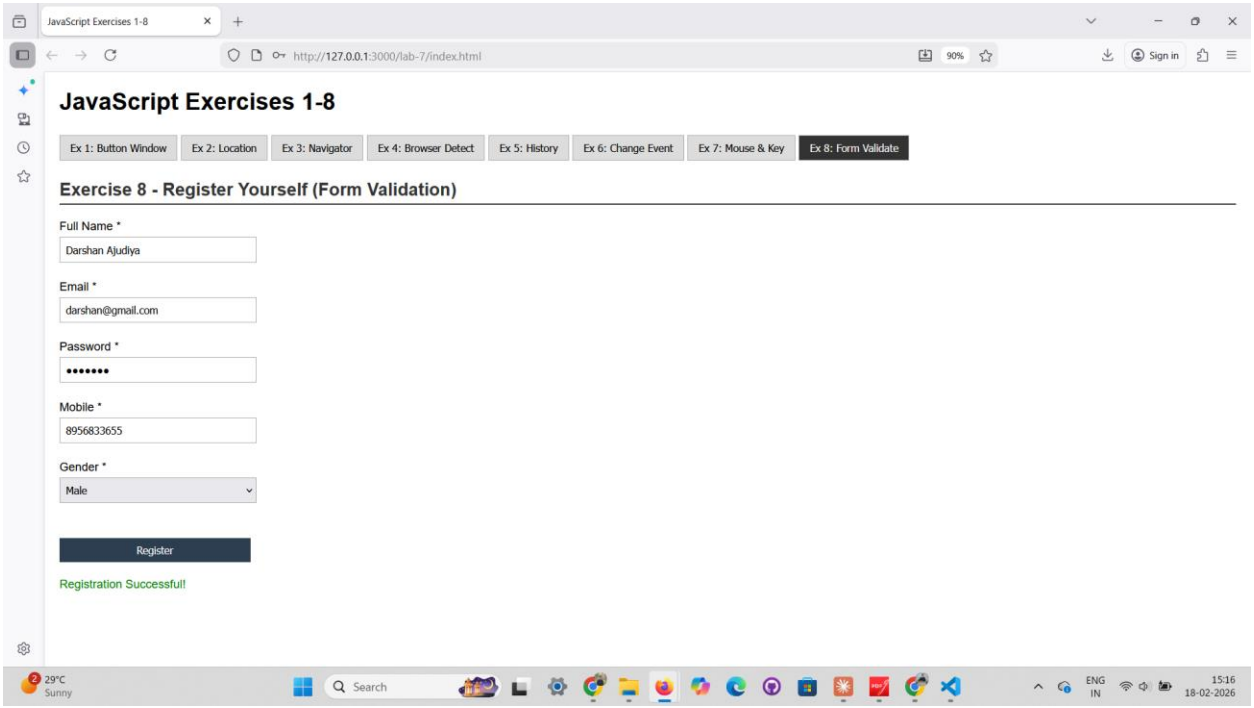
Ex 7: Mouse & Key
Ex 8: Form Validate
Exercise 7 - Mouse Location and Keyboard Key Code
Mouse Position
Move mouse here
X: 214 Y: 54
Keyboard Input
7
Key: 7
Key Code: 55
Unicode: U+0037 (55)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Ex 8: Form Validate
Exercise 8 - Register Yourself (Form Validation)
Full Name *
Darshan Ajudiya
Email *
darshan@gmail.com
Password *
Mobile *
8956833655
Gender *
Male
Register
Registration Successful!